

Sophia N. Wilson

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Research Interests

My research sits at the intersection of AI, data, and sustainability. I investigate the environmental, social, and economic costs of large-scale data practices in frontier AI, a trend I characterise as hyper-datafication. My work spans empirical assessment of these costs, technical methods for more data-frugal AI, and conceptual work related to training data governance, with the aim of translating these findings into concrete recommendations for practitioners, platforms, and policymakers.

Education

- 2025–present **PhD Fellow, Sustainable Artificial Intelligence, SAINTS Lab, Dept. of Computer Science, UCPH, Denmark.**
Resource efficient machine learning (ML), data efficient ML, data lifecycle.
- 2023–2025 **Master in Computational Physics, Niels Bohr Institute, UCPH, Denmark.**
CGPA: 12/12. Physics-informed ML, applied ML, applied statistics, scientific programming, cosmology.
- 2019–2023 **Bachelor in Physics, Niels Bohr Institute, UCPH, Denmark.**
CGPA: 11/12. Computational physics, statistics, numerical methods, astrophysics, cosmology.

Publications

In Conference Proceedings

- 2026 **Sophia N. Wilson**, Sebastian Mair, Mophat Okinyi, Erik B Dam, Janin Koch, and Raghavendra Selvan. How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. ACM, June 2026.
- 2026 **Sophia N. Wilson**, Guðrún Fjóla Guðmundsdóttir, Andrew Millard, Raghavendra Selvan, and Sebastian Mair. Position: Stop preaching and start practising data frugality for responsible development of ai. In *Proceedings of the 43rd International Conference on Machine Learning (ICML), Position Paper Track*, volume 306 of *Proceedings of Machine Learning Research*, 2026.

Preprints

- 2026 Pedram Bakhtiarifard, **Sophia N. Wilson**, Mahmoud Afifi, Jonathan Wenshøj, and Raghavendra Selvan. Characterizing learning in deep neural networks using tractable algorithmic complexity analysis, 2026.
- 2025 **Sophia N. Wilson**, Jens Hesselbjerg Christensen, and Raghavendra Selvan. Trading carbon for physics: On the resource efficiency of machine learning for spatio-temporal forecasting, 2025.

Research Projects

Dept. of Computer Science, UCPH

- Spring 2026 **Connecting Algorithmic Complexity and Deep Neural Networks.**
Collaborative work developing QuBD (Quantized Block Decomposition), a scalable method for estimating the algorithmic complexity of deep neural network weights, providing the first scalable characterisation of the learning-as-compression hypothesis for deep neural networks.
- Fall 2025 **Costs of Data Abundance and Benefits of Data Frugality in AI.**
Led two studies: (1) quantified the environmental, social, and economic costs of hyper-datafication through large-scale metadata analysis of 550,000 Hugging Face datasets, a survey of data worker conditions in Kenya, and an examination of data centre investment and geographic concentration, accepted at FAccT 2026; (2) identified a preach-practice gap in the data frugal literature where energy savings are frequently cited as motivation but rarely measured, and developed recommendations for closing it, accepted as a position paper at ICML 2026.
- Niels Bohr Institute & Dept. of Computer Science , UCPH
- Fall, 2024 – Spring, 2025 **Quantifying the Reduction in Carbon Footprint using Physics-Informed ML.**
Completed my MSc thesis applying physics-informed ML to reduce energy consumption in PDE simulations.

International Experience

- Feb. 2026 Machine Learning Summer School hosted by **Maincode, Melbourne, Australia.**
- Spring 2024 Exchange semester at the **Institute of Physics, University of Amsterdam, The Netherlands.**
- Aug. 2022 Astromatic Machine Learning Summer School at the **University of Montreal, Canada.**
- Aug. 2021 Nordic Optical Telescope Summer School **La Palma, Spain.**

Teaching Experience

- 2026 **Data Analysis and Machine Learning (for high school teachers),** *Dept. of Computer Science, Aarhus University.*
- 2025 **Machine Learning A,** *Dept. of Computer Science, UCPH.*
- 2025 **Mathematics A,** *Falkonergårdens Gymnasium.*
- 2023 **Summer Science Camp,** *Insero.*
- 2022-23 **Mathematics for Physicists 2,** *Niels Bohr Institute, UCPH.*
- 2022-23 **Mathematics for Physicists 1,** *Niels Bohr Institute, UCPH.*
- 2020-21 **Mechanics and Special Relativity,** *Niels Bohr Institute, UCPH.*

Supervision

- 2026 **Co-supervising two BSc students,** *Faculty of Humanities, UCPH.*

Invited Talks and Panel Debates

- 2026 **Speaker at P1 Workshop on Green AI,** *Pioneer Center for AI, UCPH.*
- 2026 **Speaker and panellist at Sustainability Week,** *Lund Stem Cell Center, Lund University.*
- 2025 **Speaker at Resource-Efficient AI Workshop,** *D3A Conference.*
- 2025 **Speaker at Climate Action Day,** *Copenhagen Center for Social Data Science, UCPH.*

Guest Lecturing, Workshops, and Outreach

- 2026 **Presenting Hyper-Datafication paper,** *Visual Computing, Technical University of Denmark.*
- 2026 **Presentation at AI and Sustainability Workshop by the Tech Policy Youth Committee,** *The National Center for AI in Society, UCPH.*
- 2026 **Presenting Data Frugality paper,** *Embedded AI Group, Technical University of Denmark.*

- 2026 **Co-organiser and moderator at the SAINTS'26 workshop on Sustainable AI**, *SAINTS Lab, Dept. of Computer Science, UCPH.*
- 2026 **Guest-lecturing on AI, Resource Use and Sustainability**, Københavns Professionshøjskole.
- 2026 **Organiser of a workshop on Environmental Impacts of AI**, Det Åbne Gymnasium.
- 2025 **Guest-lecturing in PhD course on Machine Learning for Sciences**, *Dept. of Computer Science, UCPH.*
- 2025 **Talks about Studying Physics at UCPH**, Falkonergårdens Gymnasium.
- 2025 **Presentation of current MSc thesis work at the 4th Annual Niels Bohr Institute MSc Student Symposium**, *Niels Bohr Institute, UCPH.*
- 2021-24 **Member of Women in Physics contributing to initiatives promoting gender balance in STEM**, *Niels Bohr Institute, UCPH.*

Referees

Raghavendra Selvan

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